

Lecture 37

4/24/26

Localization & Resolution Recap

- Finish MRI JRA
- 2 Lectures of digital image processing

(i) Select a slice

(a) z gradient pulsed on
range of frequencies about a center
sent

Location in MRI is complicated

Axes x, y, z
are arbitrary

Step 1: select a slice to be imaged,
then move to another slice

Take the slice direction to be along z

Choose frequency $\nu_c \pm \Delta\nu/2$
 $\downarrow$ center freq
 $\uparrow$ range of freq

Need to encode w/in slice to get x & y data

This is a mind-boggling process
(angle)

Step 2: phase encode along y

Pulse on a y gradient

ANIMATION

Mapping w/in slice

NOT collecting signal during phase encoding

Step 3: signal freq encode along x

Pulse on x gradient & collect signal

Doctor gets large ^{signal} frequencies where B is
big & small frequencies where B is small

Position along x distinguished by signal freq

different
angles for
different y
values
Angles distinguished
along y
position



Pick slice z
Phase encode y
Signal encode x } Look at "Encoding Summary"

Every volume element has a unique address.

This tells how each pt is addressed

How finely are addresses set? This is resolution.

Resolution is set by doctor
NOT diffraction limited

Decide volume
Decide # of slices

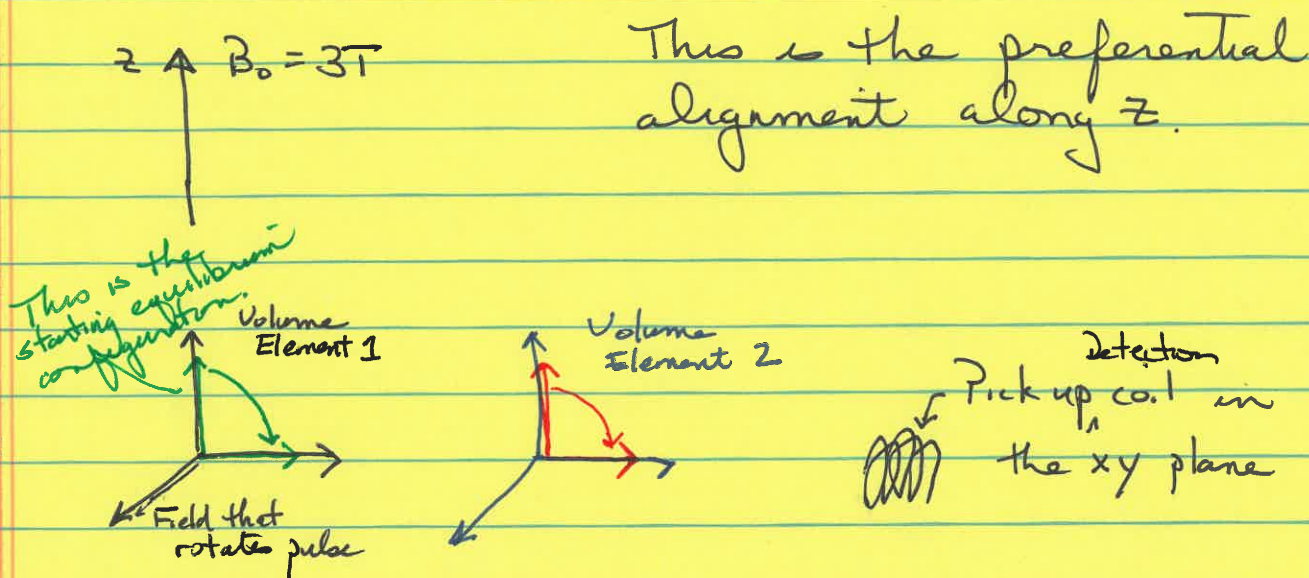
The more data you collect:

- The slower the process
- The poorer the resolution

CONTRAST

Look at "essentials"

In MRI, ^{doctor will look at different volume element w/ different contents} each ^{little} volume element in the strong field has a net magnetization M along z



Detect w/ coil

Doctor applies an RF pulse. It rotates M .

Standard pulse is a 90° pulse, which rotates magnetization into xy plane.

M_{xy} is the MRI signal
 M_z is NOT.

Start w/ equilibrium magnetization M_z
 Use 90° RF pulse to create M_{xy} (the signal)
 This is a non-equilibrium.
 M_{xy} will decay, & doctor will lose the signal
 as we return to equilibrium
 M_z will recover as we return to equilibrium

T_1 & T_2 are time scales over which
 M_{xy} & M_z recover.

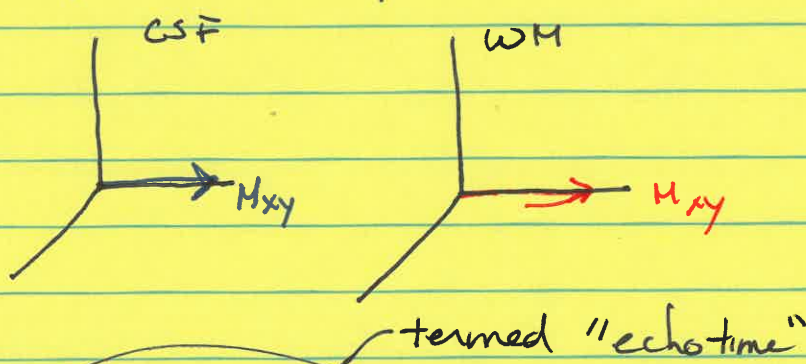
They are very different (maybe surprisingly)

T_2 is the time scale for signal decay
 $\sim 70 \text{ ms WM}$
 $\sim 280 \text{ ms CSF}$

T_1 is the time scale for recovery of M_z
 $\sim 1000 \text{ ms WM}$
 $\sim 3800 \text{ ms CSF}$

The huge differences between T_1 , T_2 for different tissues are the key to contrast in MRI.

Conceptual Example



Let ~ 150 msec elapse before collecting data.

$\Rightarrow$ WM signal is gone because $70\text{ms} < 150\text{ms}$
 Dark
 Bright CSF signal still present because $150\text{ms} < 1000\text{ms}$

Looked at

EVALUATIONS

VIDEO

- Functional MRI

Oxy/deoxy
 $\uparrow \quad \uparrow$

Different
magnetic
properties

Which parts of brain have signals correlated w/ tasks

Test = 3, 4, 5 min

Get image every 1 or 2 sec

Look at effects of injury or disease on brain

Focus NOT sharp but numerous

Anatomical - One high ^{single snap} res snapshot

Fcn - Time series looking at active
time series over

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